

Exercice :

- 1) $E = mgh$ $h = ?$
- 2) $P = f \cdot \frac{m_1 \cdot m_2}{m_3}$ $m_1 = ?$ $m_3 = ?$
- 3) $\frac{n_1}{n_2} = \frac{z_2 \cdot z_4}{z_1 \cdot z_3}$ $z_3 = ?$ $n_2 = ?$
- 4) $A = \frac{a+b}{2} \cdot h$ $a = ?$ $h = ?$
- 5) $V = \frac{\pi \cdot d^2}{4} \cdot h$ $h = ?$
- 6) $V = \frac{h}{6} \cdot (B_1 + B_2 + 4M)$ $h = ?$ $M = ?$
- 7) $D_r = \frac{D}{1 + A_v B_v}$ $D = ?$ $A_v = ?$
- 8) $P = Q \cdot \frac{R-r}{2R}$ $r = ?$
- 9) $G = \frac{kR_a}{R_i + R_a}$ $R_i = ?$
- 10) $A = \frac{F + S_a \cdot \alpha}{S_a}$ $F = ?$
- 11) $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ $R = ?$ $R_1 = ?$
- 12) $Q = \lambda \cdot \frac{T_1 - T_2}{L} \cdot A$ $T_1 = ?$
- 13) $R = \frac{\lambda L}{A} (1 + \alpha \cdot \Delta\theta)$ $A = ?$ $\alpha = ?$

Ex : page 1

1 pair de chaussure = 20
1 chaussure = 10
1 bonhomme = 5
1 paire de lunette = 2
1 gant = 20
Calcul :
 $10 + ((5 + 2 + (2 \times 20) + 20) \times 2) = 144$

Solutions :

1) $h = \frac{E}{mg}$; 2) $m_1 = \frac{P \cdot m_3}{f \cdot m_2}$; $m_3 = f \cdot \frac{m_1 \cdot m_2}{P}$; 3) $z_3 = \frac{z_2 \cdot z_4 \cdot n_2}{z_1 \cdot n_1}$; $n_2 = \frac{n_1 \cdot z_1 \cdot z_3}{z_2 \cdot z_4}$

4) $a = \frac{2A}{h} - b$; $h = \frac{2A}{a+b}$; 5) $h = \frac{4V}{\pi \cdot d^2}$; 6) $h = \frac{6V}{B_1 + B_2 + 4M}$; $M = \frac{\frac{6V}{h} - B_1 - B_2}{4}$

7) $D = D_r \cdot (1 + A_v B_v)$; $A_v = \frac{\frac{D}{D_r} - 1}{B_v}$; 8) $r = R - \frac{2PR}{Q}$; 9) $R_i = \frac{kR_a}{G} - R_a$; 10) $F = S_a \cdot (A - \alpha)$

11) $R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$; $R_1 = \frac{1}{\frac{1}{R} - \frac{1}{R_2}}$; 12) $T_1 = \frac{Q \cdot L}{A \cdot \lambda} + T_2$; 13) $A = \frac{\lambda \cdot L}{R} (1 + \alpha \cdot \Delta\theta)$; $\alpha = \frac{\frac{A \cdot R}{\lambda \cdot L} - 1}{\Delta\theta}$